

Autumn 2024

29-08-2024

MA 403 Real Analysis, Quiz 1

Duration: 45 min, Total marks: 15

1. Let x and y be real numbers such that $x < y$. Show that there exist $n \in \mathbb{N}$ such that $ny > 1 + nx$. (1.5)
2. Let S be a closed and bounded subset of $\mathbb{R}^n$. Prove that every infinite subset of S has a limit point in S . (2)
3. (a) Is every point of every open set $A \subset \mathbb{R}$, a limit point of A ? Justify. (2)
(b) Is every point of every open set A of any metric space, a limit point of A ? Justify. (2)
4. Prove that the intersection of an arbitrary collection of compact subsets of any metric space X is compact. (2)
5. Prove that a subset S of $\mathbb{R}$ is connected if it has the following property: If $x \in S, y \in S$ and $x < z < y$, then $z \in S$. (3)
6. Let $a > 0, b > 0, c > 0$ and $a + b + c = 1$. Using Cauchy-Schwarz inequality show that

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{1}{2}.$$

(2.5)

***** End of paper *****

MA 403 Real Analysis, Mid-semester Exam.

Duration: 2 Hrs, Total marks: 25

1. (a) Determine all limit points of the set $A = \{(-1)^n/[1 + (1/n)] : n \in \mathbb{N}\}$ in $\mathbb{R}$ and determine whether A is open or closed (or neither). (2)

- (b) Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_n = \sqrt{n^2 + 2} - n$. Compute $\lim_{n \rightarrow \infty} a_n$. Justify. (2.5)

2. Let X and Y be metric spaces. If $f : X \rightarrow Y$ is a continuous function and if E is a connected subset of X , then prove that $f(E)$ is connected. (3.5)

3. Let X be a compact metric space and if $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence in X , then show that $\{x_n\}_{n=1}^{\infty}$ converges to some point of X . (3)

4. State and prove Heine-Borel theorem. (3.5)

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function and assume that there exists at least one point x_0 in $\mathbb{R}$ at which f is continuous. Suppose also that, for every x and y in $\mathbb{R}$, f satisfies the equation

$$f(x + y) = f(x) + f(y).$$

Prove that there exists a constant a such that $f(x) = ax$ for all x . (3.5)

6. Either prove or disprove the following statement:
Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be such that

$$\lim_{t \rightarrow a} g(t) = b \quad \text{and} \quad \lim_{t \rightarrow b} f(t) = c.$$

Then

$$\lim_{t \rightarrow a} f(g(t)) = c. \quad (1.5)$$

[Please turn over]

Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in $\mathbb{R}$. Consider the set

$$P = \{n \in \mathbb{N} : x_n > x_m \text{ for all } m \in \mathbb{N} \text{ with } m > n\}$$

Prove the following:

- (a) If P is a finite set, then $\{x_n\}_{n=1}^{\infty}$ has a monotonically increasing subsequence.
 - (b) If P is an infinite set, then $\{x_n\}_{n=1}^{\infty}$ has a monotonically decreasing subsequence.
- (5.5)

***** End of paper *****

Autumn 2024

16-11-2024

MA 403 Real Analysis, End-semester Exam.- part 1
Duration: 1 hrs 25 min, Total marks: 22

1. (a) State completeness axiom (1)
(b) Show that a convergent sequence in a metric space has a unique limit point. (1.5) ✓
(c) Let $\{x_n\}_{n=1}^{\infty}$ be a sequence such that $\mathbb{Q} \cap [0, 1] = \{x_n : n \in \mathbb{N}\}$. Can $\{x_n\}_{n=1}^{\infty}$ be an increasing sequence? Justify. (1.5) ✓
(d) Compute the $\liminf_{n \rightarrow \infty} a_n$ and $\limsup_{n \rightarrow \infty} a_n$ where
$$a_n = 3 + (-1)^n \left(\frac{n+5}{n+6} \right)$$
 (1) ✓

2. (a) Determine all values of $\alpha \in \mathbb{R}$ for which the following series converges:
$$\sum_{n=1}^{\infty} \left(1 - n \sin \frac{1}{n} \right)^{\alpha}$$
 (2.5) $\alpha > \frac{1}{2}$

- (b) Let $a \in \mathbb{R}$. Is the series $\sum_{n=1}^{\infty} (-1)^n \sin \frac{a}{n}$ convergent? Is the series absolutely convergent? Justify. (3)

3. Prove that $\sum_{n=1}^{\infty} \frac{x}{n^{\alpha}(1+n^2)}$ converges uniformly on $\mathbb{R}$ if $\alpha > \frac{1}{2}$. (3.5)

4. Determine the set of all the points $x \in \mathbb{R}$ for which the power series converges:
$$\sum_{n=1}^{\infty} \left(\frac{2 + (-1)^n}{5 + (-1)^{n+1}} \right)^n x^n$$
 (3) $\left(-\frac{4}{3}, \frac{4}{3} \right)$

5. If $f : [a, b] \rightarrow \mathbb{R}$ is an integrable function, then prove that function $|f|$ is integrable. (5)

***** End *****

$U(p, 1/1)$

$\sin \frac{a}{n}$

By Weierstrass M-test
 $L(f) = U(f)$

$\frac{n}{2(1+n^2)}$
 $\leq \frac{1}{2n^{2+\frac{1}{2}}}$
 $\downarrow$
Cgt if $2+\frac{1}{2} > 1$
 $\therefore 2 > \frac{1}{2}$

Autumn 2024

16-11-2024

MA 403 Real Analysis,

End-semester Exam.- part 2

Duration: 1 hrs 25 min, Total marks: 23

1. Assume $f : S \rightarrow T$ is uniformly continuous on S , where S and T are metric spaces. If $\{x_n\}_{n=1}^{\infty}$ is any Cauchy sequence in S , prove that $\{f(x_n)\}_{n=1}^{\infty}$ is a Cauchy sequence in T . (3)

2. Determine all values of $s \in \mathbb{R}$ for which the following integral converges:

$$\int_0^1 x^{s-1} e^{-x} dx$$

(4)

3. (a) For a function $f : [0, 1] \rightarrow \mathbb{R}$ let $B_n(f)$ denote the Bernstein polynomial of f . For $f_2(x) = x^2$, show that

$$B_n(f_2)(x) = \left(1 - \frac{1}{n}\right)x^2 + \frac{1}{n}x$$

$$= x^2 - \frac{1}{n}x^2 + \frac{1}{n}x$$

(4.5)

- (b) State Fejer theorem.

(1.5)

4. (a) Define function of bounded variation. (1.5)

- (b) Is the function $f : [0, 1] \rightarrow \mathbb{R}$, defined as $f(x) = x^2 \sin(\frac{1}{x})$ if $x \neq 0$ and $f(0) = 0$, of bounded variation? Justify. (3.5)

5. Assume that all zeros of a polynomial P of degree $n \geq 3$ are real and write

$$P(x) = (x - a_1)(x - a_2) \dots (x - a_n),$$

where $a_i \leq a_{i+1}$, $i = 1, \dots, n-1$. Show that there exist c_i such that

$$P'(x) = n(x - c_1)(x - c_2) \dots (x - c_{n-1}),$$

where $a_i \leq c_i \leq a_{i+1}$, $i = 1, \dots, n-1$. Show that if

$$Q(x) = (x - a_1)(x - a_2) \dots (x - a_{n-1}),$$

$$Q'(x) = (n-1)(x - d_1)(x - d_2) \dots (x - d_{n-2}),$$

where $a_i \leq d_i \leq a_{i+1}$, $i = 1, \dots, n-2$, then $d_i \geq c_i$, $i = 1, \dots, n-2$. (5)

***** End *****